8.15 Solution: (a) Let us first consider the TE mode. Assume that the solution is independent of y, we can write down the longitudinal magnetic field as

$$H_z(x, y, z, t) = \psi(x)e^{i(kx-\omega t)}.$$

Then, in the two media, the wave equation becomes

$$\frac{d^2\psi(x)}{dx^2} + \left(\frac{n_i^2\omega^2}{c^2} - k^2\right)\psi(x) = 0,$$

for $i = 1, 2$. For propagating solution, we require it decays from the interface in the outer medium. Therefore, we must have

$$\gamma_1^2 = \frac{n_1^2\omega^2}{c^2} - k^2 > 0, \quad \gamma_2^2 = k^2 - \frac{n_2^2\omega^2}{c^2} > 0.$$

Since the problem here is symmetric in x, we can choose the solution to the wave equation to have definite parity. For even parity, we have

$$\psi(x) = \begin{cases} A\cos(\gamma_1 x), & |x| \le a \\ Be^{-\gamma_2|x|}, & |x| > a \end{cases}.$$

Using Eq. (8.33), we can find the transverse magnetic field in both media as

$$H_x = \frac{ik}{n^2\omega^2/c^2 - k^2}\frac{d\psi(x)}{dx} = \begin{cases} -\dfrac{ikA}{\gamma_1}\sin(\gamma_1 x), & |x| \le a \\ \\ \mathrm{sgn}(x)\dfrac{ikB}{\gamma_2}e^{-\gamma_2|x|}, & |x| > a \end{cases}.$$

The normal and tangential components of the magnetic field must be continuous at the interface of the media. This condition at the upper boundary $x = a$ indicates that

$$A\cos(\gamma_1 a) = Be^{-\gamma_2 a}, \quad \frac{A}{\gamma_1}\sin(\gamma_1 a) = -\frac{B}{\gamma_2}e^{-\gamma_2 a},$$

which is equivalent to

$$\tan(\gamma_1 a) = -\frac{\gamma_1}{\gamma_2}. \tag{1}$$

For odd parity solution, we can repeat the same calculation and find that

$$\psi(x) = \begin{cases} A\sin(\gamma_1 x), & |x| \le a \\ \mathrm{sgn}(x)Be^{-\gamma_2|x|}, & |x| > a \end{cases},$$

and

$$H_x = \frac{ik}{n^2\omega^2/c^2 - k^2}\frac{d\psi(x)}{dx} = \begin{cases} \dfrac{ikA}{\gamma_1}\sin(\gamma_1 x), & |x| \le a \\ \\ \dfrac{ikB}{\gamma_2}e^{-\gamma_2|x|}, & |x| > a \end{cases}.$$

By the boundary condition, we will obtain the following relation

$$A\sin(\gamma_1 a) = Be^{-\gamma_2 a}, \quad \frac{A}{\gamma_1}\cos(\gamma_1 a) = \frac{B}{\gamma_2}e^{-\gamma_2 a},$$

which is equivalent to

$$\cot(\gamma_1 a) = \frac{\gamma_1}{\gamma_2}. \tag{2}$$

For the TM mode, the longitudinal electric field is

$$E_z(x, y, z, t) = \psi(x)e^{i(kx-\omega t)},$$

and the even and odd parity solutions are the same as in the TE mode case, with the same expressions for the transverse electric field components. In the electric field case, the continuity condition becomes

$$E_z(x = a-) = E_z(x = a+), \quad \epsilon_1 E_x(x = a-) = \epsilon_2 E_x(a+),$$

which translates to

$$A\cos(\gamma_1 a) = Be^{-\gamma_2 a}, \quad \frac{n_1^2 A}{\gamma_1}\sin(\gamma_1 a) = -\frac{n_2^2 B}{\gamma_2}e^{-\gamma_2 a},$$

for even parity, and

$$A\sin(\gamma_1 a) = Be^{-\gamma_2 a}, \quad \frac{n_1^2 A}{\gamma_1}\cos(\gamma_1 a) = \frac{n_2^2 B}{\gamma_2}e^{-\gamma_2 a},$$

for odd parity, where we have used $\epsilon_i = n_i^2$. Then, we can obtain similar relations as in the TE mode case,

$$\tan(\gamma_1 a) = -\frac{n_2^2}{n_1^2}\frac{\gamma_1}{\gamma_2}, \tag{3}$$

for even parity, and

$$\cot(\gamma_1 a) = \frac{n_2^2}{n_1^2}\frac{\gamma_1}{\gamma_2}, \tag{4}$$

for odd parity.

Introduce the fiber parameters,

$$V\xi = \frac{n_1\omega}{c}a\sin\theta = \gamma_1 a, \quad \xi^2 = \frac{\gamma_1^2}{n_1^2\omega^2/c^2}\frac{n_1^2}{n_1^2 - n_2^2} = \frac{\gamma_1^2}{\gamma_1^2 + \gamma_2^2}, \quad \sqrt{\frac{1}{\xi^2} - 1} = \frac{\gamma_2}{\gamma_1},$$

we can write Eqs. (1-2), for TE mode, as

$$\cot(V\xi) = -\sqrt{\frac{1}{\xi^2} - 1}, \quad \tan(V\xi) = \sqrt{\frac{1}{\xi^2} - 1}.$$

Notice that

$$\tan\left(x - \frac{p\pi}{2}\right) = \begin{cases} \tan x, \ p \text{ even} \\ -\cot x, \ p \text{ odd} \end{cases},$$

we have

$$\tan\left(V\xi - \frac{p\pi}{2}\right) = \sqrt{\frac{1}{\xi^2} - 1}. \tag{5}$$

Similarly, for TM mode, Eqs. (3-4), we will have

$$\tan\left(V\xi - \frac{p\pi}{2}\right) = \frac{n_1^2}{n_2^2}\sqrt{\frac{1}{\xi^2} - 1}. \tag{6}$$

If we define $\Delta = (n_1^2 - n_2^2)/2n_1^2$, and $f = 1$ for TE mode and $f = 1/(1-2\Delta)$ for TM mode, we can then combine Eqs. (5-6) into one equation,

$$\tan\left(V\xi - \frac{p\pi}{2}\right) = f\sqrt{\frac{1}{\xi^2} - 1}.$$

(b) For TE mode, we can write Eq. (5) as

$$V\xi = \frac{p\pi}{2} + \arctan\left(\sqrt{\frac{1}{\xi^2} - 1}\right)$$

For $V \gg 1$ and small p, $\xi \ll 1$. Using the asymptotic expansion of the arctan function at infinity,

$$\arctan x = \frac{\pi}{2} - \frac{1}{x} + \frac{1}{3x^2},$$

we have

$$V\xi = \frac{p\pi}{2} + \frac{\pi}{2} - \frac{\xi}{1-\xi^2} + \frac{\xi^3}{3(1-\xi^2)^{3/2}} = \frac{(p+1)\pi}{2} - \xi - \frac{\xi^3}{6}.$$

We can formally express the solution to this equation as

$$\xi = \frac{(p+1)\pi}{2(V+1)}\left[1 + \frac{\xi^2}{6(V+1)}\right]^{-1}.$$

Since ξ is small, we use its lowest order solution

$$\xi_0 = \frac{(p+1)\pi}{2(V+1)}$$

to approximate the right hand side to obtain

$$\xi = \frac{(p+1)\pi}{2(V+1)}\left[1 + \frac{\xi_0^2}{6(V+1)}\right]^{-1} = \frac{(p+1)\pi}{2(V+1)}\left[1 - \frac{\xi_0^2}{6(V+1)}\right] = \frac{(p+1)\pi}{2(V+1)}\left[1 - \frac{(p+1)^2\pi^2}{24(V+1)^3}\right].$$